

Statement of Work — Production PCB (Transcription + Layout)

One 4-layer board · execution-only · fixed-price

What this project is

A consumer voice-assistant device, going to a 1,000-unit production run assembled from JLCPCB turnkey PCBA (pre-assembled, pre-flashed). The system is **already designed and validated**: it runs end-to-end on an ESP32-S3 devkit with breakout boards, every production IC has been selected and verified in JLC/LCSC stock, and every GPIO assignment is proven in working firmware.

Your job is the remaining EE execution for the main board only:

1. **Production schematic** — transcribe the provided, validated devkit-level schematic onto the bare production parts (component mapping table provided: devkit → module, breakout → IC, every part with an LCSC number), using each part's **public datasheet reference circuit**. No part research, no architecture decisions.
2. **4-layer SMT layout** — to the physical spec in [03-physical-spec.md](#) (target envelope 60 × 50 mm; may grow to 60 × 60 mm or a sideways-T, ≤60 × 80 mm bbox, if EMC/density needs it; defined connector zones + two-zone height limits; you place mounting holes freely).
3. **JLC-ready outputs** — Gerbers + BOM + CPL for direct JLCPCB upload.
4. **One revision** included after the client's prototype bring-up.

What this project is NOT

- **No firmware** — none changes hands in either direction. The TAS5805M's I2C register init, OTA, audio pipeline, etc. are all client-side and happen after delivery. You never need to run, read, or write any code.
- **No part selection** — every IC, connector, and module is locked and listed with LCSC numbers in [02-technical-spec.md](#). (You choose passive values per the reference circuits; prefer JLC Basic library.)
- **No daughterboards** — three small auxiliary boards exist in the product (controls, microphone, glow-LED); the client designs those. Your board just carries their JST headers.
- **No bring-up, validation, or test fixtures** — client's job, after boards arrive.
- **No enclosure work** — the client is the mechanical engineer with in-house printing. You get the envelope + a few placement zones; the enclosure adapts to your layout.
- FCC test logistics / SDoC / labeling are **client-side** — but the board **must be designed to pass FCC Part 15B** (see [02-technical-spec.md](#) §6). No production order management.

Provided to you at kickoff (the complete package)

File	Contents
02-technical-spec.md	Component mapping (prototype part → production part + LCSC #), full GPIO pin map, per-block circuit requirements, connector pinouts, production essentials, layout rules

File	Contents
03-physical-spec.md	Board envelope, mounting-hole policy, placement zones, height limits
04-schematic/	The validated devkit-level schematic (PDF, GPIO-annotated, + source)
reference/	All datasheets + the TI EVM user guide + Espressif hardware design guidelines (PDFs — nothing to hunt down)

Deliverables

Deliverable	Format
Editable source project (KiCad or EasyEDA Pro)	full project, work-for-hire transfer
Schematic PDF	single reviewable PDF
Assembly drawing	PDF, top + bottom with refdes
BOM	CSV/XLSX: refdes, qty, MPN, LCSC #, Basic/Extended flag
CPL (centroid)	CSV, JLC format
Gerbers + drill	RS-274X + Excellon, JLC naming, upload-ready
Clean DRC/ERC confirmation	note in delivery email is fine
One post-prototype revision	updated files, included in fixed price

What this board involves (so the scope is clear)

- A 4-layer mixed-signal layout: ESP32-S3-WROOM module (**antenna keepout**), TAS5805M Class-D amp (**PowerPAD thermal + LC output filter, mono PBTL**), 2× TPS54331 bucks, 2× logic level shifters, P-FET/PTC input protection, and JST headers.
- Tool: **KiCad or EasyEDA Pro** — whichever you work in; you deliver the editable source (work-for-hire) plus the JLC outputs above.

Process & terms

- **Scope is locked and fixed-price** — this is a one-board transcription + layout + production-prep job; design, validation, and sourcing are already done. Hourly only if capped and agreed.
- **Timeline:** ~2–4 weeks to first Gerber drop. Revision turnaround 1–2 weeks after client bring-up.
- **Review:** one async schematic/layout review pass with the client before fabrication (marked-up PDF is fine).
- **Milestones / payment:** 25% kickoff / 50% Gerber drop / 25% revision sign-off.
- **IP:** work-for-hire; all deliverables fully assigned to the client. Portfolio use permitted.